

Swiftflitz Car Rental Management System

Technical Requirements Document

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Project Name:	Swiftflitz Car Rental Web Application
Technology Stack:	Laravel (Backend), Modern Web Frontend
Document Type:	Functional Requirements Specification
Prepared For:	Swiftflitz Management Team

1. PROJECT OVERVIEW

Swiftflitz requires a comprehensive web-based car rental management system that enables efficient fleet management, online bookings, and operational oversight. The system will consist of two primary components: a public-facing website for customer interactions and an administrative panel for internal operations management.

2. PUBLIC WEBSITE REQUIREMENTS

2.1 Vehicle Display & Information

- Display comprehensive vehicle catalog with high-quality images
- Show make and model information prominently for each vehicle
- Display vehicle specifications (seating capacity, transmission type, fuel type, etc.)
- Show vehicle features and amenities
- Display rental pricing (only for vehicles with pricing enabled by admin)
- Show vehicle availability status in real-time
- Display vehicle categories/classes (economy, luxury, SUV, etc.)

2.2 Search & Filter Functionality

- Advanced search with multiple filter options:
 - Vehicle type/category
 - Price range (when applicable)
 - Seating capacity
 - Transmission type (automatic/manual)
 - Availability dates
 - Features (GPS, child seat, etc.)
- Real-time filter results without page reload
- Sort options (price low-to-high, newest vehicles, popular, etc.)

2.3 Booking System - Direct Online Booking (Price Enabled)

- Select pickup date and time
- Select return date and time (or specify rental duration in days)
- Calculate total rental cost automatically based on duration
- Airport pickup/drop-off option with location selection

- Customer information form:
 - Full name
 - Contact number
 - Email address
 - Driver's license details
 - Additional notes/requests
- Booking confirmation page with reference number
- Automated email confirmation to customer
- WhatsApp notification to administrator

2.4 Reservation System - Quote Request (Price Disabled)

- Vehicle reservation form for cars without public pricing
- Collect same customer details as direct booking
- Select desired pickup/return dates and times
- Airport pickup option selection
- Additional requirements/preferences field
- Submit reservation request to admin via:
 - WhatsApp message with all details
 - Admin panel notification
 - Email notification
- Confirmation message to customer that request has been received

2.5 Additional Website Features

- About Us page with company information
- Contact page with form, phone, email, and location
- Terms and conditions page
- Privacy policy page
- FAQ section for common customer questions
- Rental policies and guidelines

- Customer testimonials/reviews section
- Newsletter subscription option
- Social media integration
- Responsive design for mobile, tablet, and desktop
- SEO-optimized content and structure

3. ADMIN PANEL REQUIREMENTS

3.1 User Management & Access Control

Roles & Permissions

System Roles:

Administrator

Manager

Custom Roles (extensible)

- Role-based access control (RBAC)
- Ability to create custom roles with specific permissions
- Granular permission settings for different system modules
- User management interface for adding/editing/deactivating users
- Activity logging per user for accountability
- Secure authentication with password requirements
- Password reset functionality

3.2 Vehicle Management

Add New Vehicle

- Vehicle basic information:
 - Make and model
 - Year of manufacture
 - License plate number
 - VIN (Vehicle Identification Number)
 - Color
 - Mileage
- Vehicle specifications:
 - Category/class
 - Seating capacity
 - Transmission type

- Fuel type
- Engine size
- Vehicle features (checkboxes for GPS, Bluetooth, child seats, etc.)
- Multiple image upload for vehicle gallery
- Rental pricing configuration:
 - Daily rate
 - Weekly rate (optional)
 - Monthly rate (optional)
 - Toggle to show/hide price on public website
 - Note: Price always visible on admin side regardless of toggle
- Vehicle condition notes
- Insurance information
- Maintenance schedule/history

Edit/Update Vehicle

- Modify all vehicle information fields
- Update pricing and pricing visibility
- Add/remove vehicle images
- Update availability status
- Add maintenance records

Vehicle Availability Management

- Manually mark vehicle as available/unavailable
- Override availability for vehicle swaps (admin can make rented car available)
- View vehicle availability calendar
- Note: All vehicles without price tags are assumed available

3.3 Rental/Booking Management

Create New Rental

- Select vehicle from available inventory
- Enter customer information:
 - Full name
 - Contact number
 - Email address
 - Driver's license details (number, expiry date)
 - ID/Passport number
 - Address
- Rental details:
 - Pickup date and time
 - Expected return date and time
 - Rental duration (calculated automatically)
 - Pickup location (including airport pickup option)
 - Drop-off location
- Pricing and payment:
 - Rental rate (auto-populated from vehicle)
 - Total cost calculation
 - Additional charges (insurance, extras, etc.)
 - Deposit amount
 - Payment method
 - Payment status
- Vehicle condition documentation at pickup:
 - Upload video of vehicle condition before rental
 - Upload photos of vehicle exterior/interior
 - Fuel level at pickup
 - Existing damage notes
 - Current mileage reading
- Record which manager is processing the rental

- Additional notes and special instructions
- Generate rental agreement/contract document

Rental Status Management

Status	Description
Pending	Booking created but not yet confirmed or vehicle not yet picked up
Rented	Vehicle currently with customer during active rental period
Returned	Vehicle returned by customer but pending admin approval
Completed	Rental approved and closed by admin after return inspection
Overdue	Vehicle not returned by expected return date/time

Vehicle Return Process

- Mark vehicle as returned (changes status from Rented to Returned)
- Admin/Manager approval required to complete return:
 - Inspect vehicle condition
 - Record return mileage
 - Record fuel level at return
 - Compare with pickup condition documentation
- Return approval form includes:
 - Comment section (required):
 - "Car returned in good condition" option
 - "Car damaged" option with description field
 - General notes field
 - Damage documentation (if applicable):

- Upload multiple photos of damaged parts
- Description of each damage
- Estimated repair cost
- Additional charges applied (cleaning fees, damage fees, late fees, etc.)
- Upon approval, status changes from Returned to Completed
- Vehicle automatically becomes available again in the system
- Generate return inspection report

3.4 Dashboard & Reporting

Dashboard Overview

- Key metrics at a glance:
 - Total vehicles in fleet
 - Currently rented vehicles
 - Available vehicles
 - Overdue rentals
 - Pending returns
 - Revenue statistics (daily, weekly, monthly)
- Recent activity feed
- Upcoming returns calendar view
- Overdue rentals alert section
- Quick action buttons (add vehicle, create rental, etc.)

Reporting & Analytics

- Review all bookings by manager:
 - Filter by manager name
 - Date range selection
 - View total bookings per manager
 - Revenue generated per manager
- Vehicle utilization reports:

- Rental frequency per vehicle
- Revenue per vehicle
- Downtime analysis
- Financial reports:
 - Revenue by time period
 - Outstanding payments
 - Deposit tracking
 - Damage/repair costs
- Customer reports:
 - Customer rental history
 - Repeat customers
 - Customer payment status
- Export reports to PDF and Excel

3.5 Additional Admin Features

Notifications & Alerts

- Real-time notifications for:
 - New online bookings
 - New reservation requests (from price-disabled vehicles)
 - Upcoming returns (24 hours before)
 - Overdue rentals
 - Vehicles requiring maintenance
- WhatsApp integration for critical notifications
- Email notifications
- In-app notification center

System Settings

- Company information management
- WhatsApp API configuration

- Email templates customization
- Rental terms and conditions editor
- Tax and fees configuration
- Business hours settings
- Airport locations management
- Currency settings

Customer Management

- Customer database with full rental history
- Customer profile management
- Document storage (license copies, ID copies)
- Customer notes and flags (VIP, blacklist, etc.)
- Search and filter customers

4. TECHNICAL REQUIREMENTS

4.1 Technology Stack

- Backend Framework: Laravel (latest stable version)
- Database: MySQL/PostgreSQL
- Frontend: Modern responsive framework (Vue.js, React, or Blade templates)
- File Storage: Support for video and image uploads
- API Integration: WhatsApp Business API

4.2 Security Requirements

- Secure user authentication and session management
- Password encryption using industry standards
- CSRF protection
- SQL injection prevention
- XSS protection
- Secure file upload validation
- HTTPS enforcement
- Regular security updates and patches

4.3 Performance Requirements

- Fast page load times (under 3 seconds)
- Optimized database queries
- Image optimization and lazy loading
- Caching implementation for frequently accessed data
- Responsive design for all device sizes

4.4 Data Management

- Regular automated backups

- Data export functionality
- Audit trail for critical operations
- Soft delete implementation for data recovery

5. INTEGRATION REQUIREMENTS

- WhatsApp Business API for notifications and booking requests
- Email service provider (SMTP configuration)
- Payment gateway integration
- Google Maps API for location services
- SMS gateway for customer notifications (optional)

Swiftflitz Car Rental Management System - Requirements Document

This document is confidential and intended solely for Swiftflitz management review